

AMBUD LAHIRI

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EDUCATION

Bachelor of Technology in Computer Science (Artificial Intelligence)

Bennett University

August 2022 – Present

CGPA : 9.45

EXPERIENCE

Undergraduate Research Intern

Bennett University

August 2024 – Present

- Engineered a hypervisor-level cache side-channel TLB attack detection system using Intel's PCM, achieving 85% accuracy in identifying Flush+Reload, Flush+Flush, and Prime+Probe attacks through real-time monitoring of IPC.
- Added Explainable AI using SHAP to analyze feature importance by leveraging GradientExplainer.

App Development Lead

Syntalix

January 2024 – July 2024

- Piloted the development of a music and video streaming app utilizing Flutter and Dart, decreasing app latency by 60% and improving UI responsiveness through code optimization.
- Reduced app development cycle time by 25% through the introduction of daily stand-up meetings and sprint retrospectives, addressing critical roadblocks and fostering continuous improvement within the team.
- Conducted 50+ user interviews and usability tests throughout the app development lifecycle, directly influencing design decisions and resulting in 90% positive user feedback on core features.

PROJECTS

Harmonix | Flutter, Dart, Firebase

November 2024 – December 2024

- Engineered a Flutter app delivering both music and video content, achieving 92% operational efficiency on Android devices.
- Crafted three distinct interfaces—a secure login portal, a personalized account hub, and a comprehensive media dashboard thereby reducing onboarding duration by 28%.
- Launched interactive playlist creation feature using Flutter, enabling users to personalize content experiences, which resulted in a substantial 35% growth in net user engagement metrics on the app.
- Refined the visual experience with a striking gradient theme and custom media controls, resulting in a 25% increase in overall user retention.

BU-GPT | Flutter, Dart, OpenAI API, Firebase

August 2024 – September 2024

- Developed a Flutter-based mobile application integrated with the OpenAI API, providing interactive conversational AI and real-time document query functionalities, resulting in a 90% reduction in time spent on manual document analysis.
- Fortified mobile app security via Firebase Authentication, constructing an intuitive tab view navigation that streamlined user onboarding, decreasing the time spent by new users by 70%.
- Leveraged RESTful APIs and optimized asynchronous request handling using Dart's async/await, delivering responsive AI-generated insights, thereby improving user productivity by up to 95%.
- Revolutionized user experience through intuitive UI/UX design, leading to a 60% increase in daily active users and a 90% reduction in user-reported accessibility issues.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL (MySQL), NoSQL(MongoDB), Dart, JavaScript, Typescript, HTML, CSS.

Frameworks: Flutter, Next.js, Qt.

Developer Tools: Git, GitBash, VS Code, Visual Studio, PyCharm, IntelliJ, Android Studio, Firebase, Tkinter.

CERTIFICATIONS

- Algorithmic Toolbox
- Supervised Machine Learning
- Software Development Processes and Methodologies